

SAFETY DATA SHEET

BRSCMC® — Sodium Carboxymethyl Cellulose (CMC)

Prepared in accordance with the Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

Section 1: Identification

Product identifier: BRSCMC®

Chemical name: Sodium Carboxymethyl Cellulose (CMC), oil drilling grade

Recommended use: Viscosifier, flow-control agent, and filtration reducer for drilling fluids and completion fluids; may replace guar gum in fracturing fluids

Manufacturer / Supplier: Brighter Star Drilling Fluids

Website: www.pcenigeria.com

Section 2: Hazards Identification

Hazard classification: Not classified as a hazardous substance under GHS / OSHA Hazard Communication Standards. The following minor hazards exist in practice

- Dust irritation — inhalation, skin contact, or eye contact with CMC powder may cause mild irritation to the respiratory tract, skin, and eyes; ingestion may irritate the digestive tract
- Combustible dust — fine CMC powder dispersed in air may form a combustible dust cloud; ignition sources may cause flash fire or dust explosion
- Combustibility — combustible at elevated temperatures; autoignition temperature of the dust is approximately 370 °C
- Static hazard — static charges generated during package emptying or pneumatic transfer may trigger flash fire

Section 3: Composition / Information on Ingredients

Chemical characterization: Substance.

Component	CAS No.
Sodium carboxymethyl cellulose (Cellulose Gum)	9004-32-4

Section 4: First-Aid Measures

Exposure Route	Action
Eye contact	Rinse immediately with plenty of clean water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Remove contact lenses if

Exposure Route	Action
	present. Seek medical attention if irritation persists
Skin contact	Remove contaminated clothing and shoes. Wash the affected area thoroughly with soap and plenty of water. Launder clothing before reuse. Seek medical advice if irritation develops or persists
Inhalation	Move the victim to fresh air immediately and keep at rest in a comfortable breathing position. If breathing is difficult, administer oxygen; if not breathing, give artificial respiration. Seek medical attention
Ingestion	Do NOT induce vomiting. Rinse mouth thoroughly with water. If conscious and alert, drink 2–4 cupfuls of water or milk. Never give anything by mouth to an unconscious person. Seek medical attention

Note to physician: Treat symptomatically and supportively. No specific antidote is known.

Section 5: Fire-Fighting Measures

Suitable extinguishing media: Water spray, dry chemical, carbon dioxide (CO₂), or foam

Unsuitable media: High-pressure water jets

Autoignition temperature: ≈ 370 °C (dust)

NFPA rating: Health 0 · Flammability 1 · Instability 0

Hazardous combustion products: Carbon monoxide (CO), carbon dioxide (CO₂), and potentially other irritant or toxic fumes from thermal decomposition

Protection of firefighters: Wear self-contained breathing apparatus (SCBA, MSHA/NIOSH-approved or equivalent) in pressure-demand mode and full protective gear. Fight fire from upwind

Precaution: Avoid stirring up dust during fire-fighting to prevent dust explosion

Section 6: Accidental Release Measures

- Isolate the spill area and keep unauthorized personnel away
- Ventilate the area and wear appropriate PPE (dust mask, gloves, goggles, protective clothing)
- Avoid generating dust — dry sweeping is not recommended. Use a vacuum with a HEPA filter to collect the powder, or pre-dampen with water before sweeping
- Collect the material into clean, sealed containers for recycling or disposal
- Prevent the product from entering drains, sewers, surface water, or groundwater
- Clean the affected surface with water or alcohol as appropriate
- Dispose of contaminated material in accordance with local environmental regulations

Section 7: Handling and Storage

Handling: Avoid dust formation at all times and provide adequate local exhaust ventilation where dust may be generated. Do not breathe dust; avoid contact with eyes, skin, and clothing. Ground and bond all

equipment, containers, and transfer lines. Do not empty packages near flammable vapors or ignition sources. Handle gently using scoops, funnels, or spoons; avoid shaking or vigorous stirring that disperses dust. Prohibit eating, drinking, and smoking in handling areas; wash hands thoroughly with soap and water after handling. Keep containers tightly closed when not in use. Access limited to trained personnel only

Storage: Store in a cool, dry, clean, and well-ventilated place away from direct sunlight, at 5–40 °C and relative humidity $\leq 80\%$ (ideally $< 50\%$). CMC is hygroscopic — moisture ingress causes caking and viscosity loss. Keep containers tightly closed; after opening, expel excess air from the inner PE bag and reseal immediately. Segregate from strong acids, strong alkalis, and strong oxidizing agents. Keep away from heat, ignition sources, and flammable substances; no smoking in storage areas. Stack bags on pallets off the ground and do not stack excessively high. Ground all bulk storage and transfer equipment. Do not store together with toxic, harmful, or odor-emitting substances; keep away from food, feed, and drinking water

Shelf life: 24 months from the date of manufacture under proper sealed-storage conditions

Section 8: Exposure Controls / Personal Protection

Protection	Recommended Measures
Eyes	Chemical safety goggles compliant with OSHA 29 CFR 1910.133 or EN 166; full goggles recommended in dusty environments
Skin / hands	Impervious rubber gloves (nitrile or latex preferred); protective clothing or aprons; long sleeves
Respiratory	NIOSH/MSHA-approved dust mask or respirator (EN 149 P1/P2 or equivalent) when dust levels are significant or ventilation is inadequate
Footwear	Safety shoes
Facilities	Emergency eyewash fountain and safety shower must be accessible in the immediate work area

Hygiene measures: Wash hands thoroughly with soap and water after handling and before eating or drinking. Launder contaminated clothing before reuse.

Section 9: Physical and Chemical Properties

Property	Value
Appearance	White or slightly yellowish free-flowing powder or granules
Odor	Odorless
Taste	Tasteless
Solubility	Readily soluble in water, forming a colloidal solution; insoluble in ethanol, ether, acetone, and other organic solvents

Property	Value
Autoignition temperature	≈ 370 °C (dust)
Hygroscopicity	Hygroscopic — cakes and loses viscosity on moisture ingress

Section 10: Stability and Reactivity

Chemical stability: Stable under recommended storage conditions

Incompatible materials: Strong acids, strong alkalis, and strong oxidizing agents

Conditions to avoid: Dust formation, accumulation of static charge, heat, ignition sources, and moisture

Hazardous decomposition products: Carbon monoxide (CO) and carbon dioxide (CO₂) from thermal decomposition

Hazardous reactions: Fine dust dispersed in air may form explosive dust–air mixtures

Section 11: Toxicological Information

Acute toxicity: Non-toxic; not classified

Irritation: Dust may cause mild irritation to the respiratory tract, skin, and eyes; ingestion may irritate the digestive tract

Medical treatment: Treat symptomatically and supportively; no specific antidote is known

Section 12: Ecological Information

Biodegradability: Biodegradable

Environmental precautions: Prevent the product from entering drains, sewers, surface water, or groundwater; spills must be contained and properly disposed of

Section 13: Disposal Considerations

- Dispose of the product and its container in accordance with local, regional, and national regulations
- Uncontaminated residual product may be landfilled where permitted
- Do NOT discharge CMC powder or slurry into drains, sewers, or waterways
- Empty containers may pose a dust explosion risk if residual powder is present — clean thoroughly before disposal or recycling

Section 14: Transport Information

Not classified as a hazardous substance; not regulated as a dangerous good for transport.

Section 15: Regulatory Information

GHS / OSHA HCS: Not classified as a hazardous substance

NFPA 704: Health 0 · Flammability 1 · Instability 0

Section 16: Other Information

Remark: The above safety precautions are compiled from industry-standard SDS references for Sodium Carboxymethyl Cellulose. For formal export documentation, regulatory submission, or batch-specific compliance, always refer to the official SDS issued by the manufacturer for the exact grade and batch — parameters such as degree of substitution, viscosity, and impurity profile may vary between grades and manufacturers.

Disclaimer: The information contained in this Safety Data Sheet is provided for reference only and is based on the current state of knowledge. It describes the product with regard to safety requirements and does not guarantee any specific properties. In all cases, the actual conditions of use shall prevail. To the fullest extent permitted by applicable law, Brighter Star Drilling Fluids assumes no responsibility or liability for any results, losses, or damages arising from the use of, or reliance on, this document.